

PROJECT II — STABILITY & INTEGRITY OF AEROSPACE STRUCTURES

Work Distribution & MATLAB Implementation Guide

Aircraft: Airbus A330-200 | UC3M – Aerospace Structures

■ Required Inputs from Project I (A330-200)

The following quantities computed in **Project I** for the **Airbus A330-200** must be retrieved and handed to Project II. Three cross-sections are analysed: **root (sec 1)**, **kink (sec 2)** and **tip (sec 3)**.

Parameter	Symbol	Unit	Notes
Bending moment (pull-up, $n_{z\leq 2.5}$)	$M(y)$	N·mm	At each of the 3 sections
Shear force (pull-up)	$V(y)$	N	At each section
Wing chord at section	c	mm	Lightest wing-box from Proj. I
Wing-box width	w	mm	Fraction of chord ($\alpha \cdot c$)
Max airfoil thickness	d	mm	At section spanwise location
Airfoil t/c ratio	t/c	–	For A330-200 airfoil
Section span position	y	mm	Root / kink / tip stations
Wing semi-span	$b/2$	mm	Total semi-span of A330-200
Skin thicknesses (existing)	t_{skin}	mm	From lightest design
Spar web thicknesses	t_{web}	mm	Front & rear spar

A330-200 reference values used for geometry: semi-span ≈ 30.9 m, root chord ≈ 11.0 m, kink chord ≈ 5.8 m (at ~ 35 % span), tip chord ≈ 2.7 m; $t/c \approx 0.15$ at root $\rightarrow 0.11$ at tip. Rib spacing $L \approx 720$ mm (≈ 28.3 in) consistent with commercial aircraft data.

■ Work Distribution — 5 Members

Tasks are divided so that each person carries a **comparable workload** (~ 20 % of total effort). Shared tasks are indicated with ★. The estimated effort in brackets is qualitative (S=Small, M=Medium, L=Large).

PERSON 1 Preliminaries & Geometry		
Sub-task	Description	Effort
Task 1-I	Retrieve lightest wing-box from Project I (M , V , geometry for 3 sections)	M
Task 1-II	Rectangular wing-box simplification: compute h_b , b_W , z_{CG} per section	M
★ Shared	Compile common input parameter file for MATLAB (N , w , h_b per section)	S
Task 1-III	Material selection justification & Ramberg-Osgood curve plots for all 4 materials	S
<i>Deliverable: Geometry tables, $N(y)$ values, R-O curves report section</i>		

PERSON 2 Variable-b Sizing — Z Stringers		
Sub-task	Description	Effort
Task 2-I (a)	MATLAB: plasticity-correction + N-cycle iteration for Z-stringer, variable b	L
Task 2-I (b)	Read structural efficiency charts; implement α_{opt} search for all 3 sections	M
Task 2-I (c)	Compute panel geometry: b , t , t_s , h , z_{CG} , b_W for Z-stringer, var-b, 7075	M

Sub-task	Description	Effort
★ Shared	Coordinate sigma_opt results with Person 3 for comparison table	S
<i>Deliverable: Z-stringer variable-b results table + cross-section sketches</i>		

PERSON 3Variable-b Sizing — Integral Stringers		
Sub-task	Description	Effort
Task 2-II (a)	MATLAB: plasticity + N-cycle iteration for integral stringer, variable b	L
Task 2-II (b)	Implement Catchpole structural efficiency for integral stringer, all sections	M
Task 2-II (c)	Compute panel geometry for integral stringer, var-b, 7075	M
★ Shared	Weight calculation for Tasks 2-I & 2-II; compare Z vs integral (variable b)	S
<i>Deliverable: Integral-stringer variable-b results + weight comparison (Z vs integral)</i>		

PERSON 4Constant-b Sizing (Manufacturing Constraints)		
Sub-task	Description	Effort
Task 3-I	Select stringer pitch b from A330-200 data; justify choice	S
Task 3-II	MATLAB: alpha_b cycle for Z-stringer with fixed b, all 3 sections	L
Task 3-III	MATLAB: alpha_b cycle for integral stringer with fixed b, all sections	L
Task 3-IV	Weight comparison: variable-b vs constant-b for both stringer types	M
<i>Deliverable: Constant-b results tables, manufacturing constraint discussion, weight delta</i>		

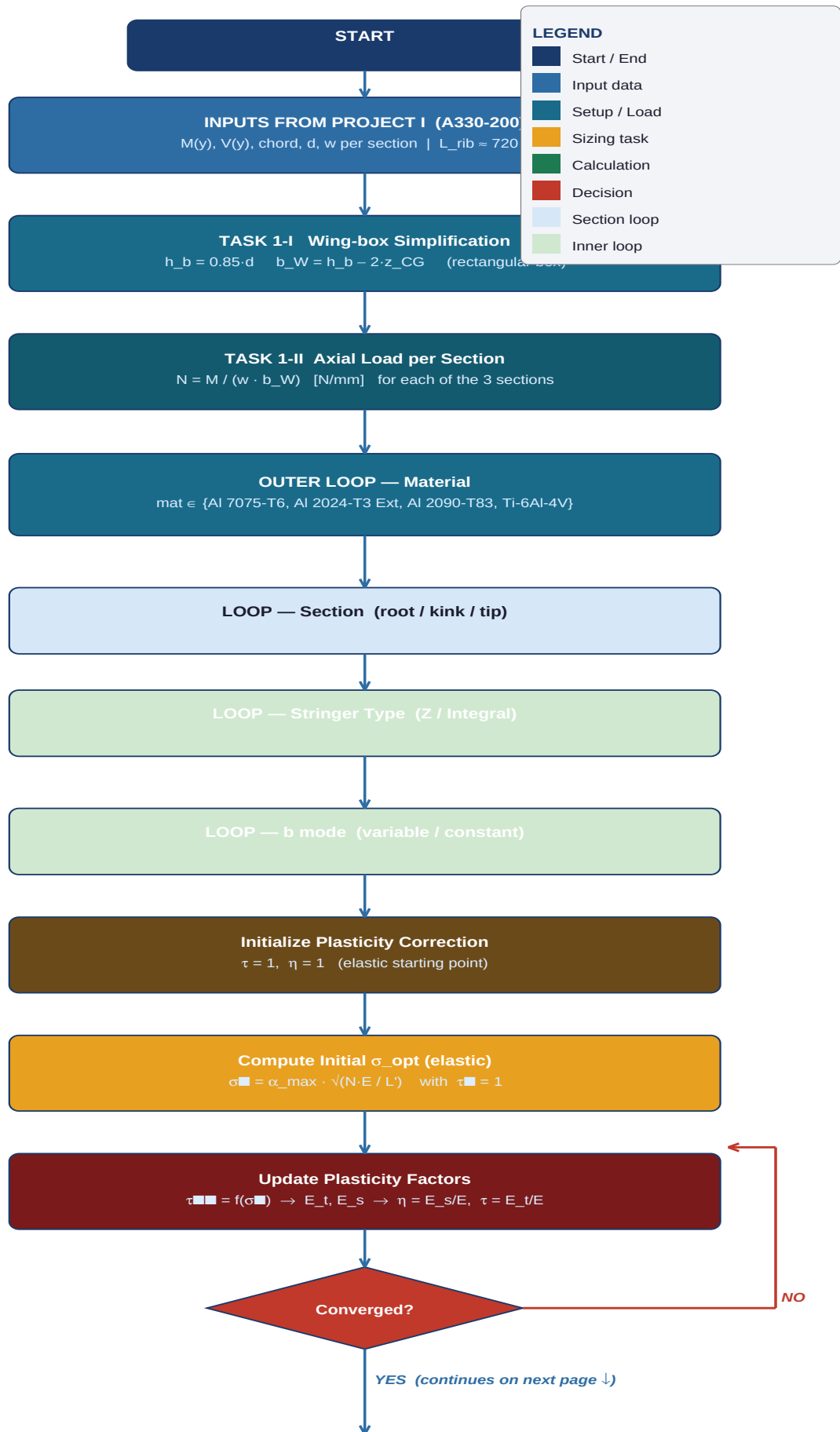
PERSON 5Material Sensitivity & Integration		
Sub-task	Description	Effort
Task 4-I	Re-run variable-b & constant-b sizing for 2024-T3 Ext, 2090-T83, Ti-6Al-4V (root & tip)	L
Task 4-II	Produce material comparison: W vs material for both stringer types & b conditions	M
★ Shared	Assemble final report: merge sections from all persons, check consistency	M
★ Shared	Produce executive summary, conclusions & recommendations	S
<i>Deliverable: Material sensitivity plots, final merged report, conclusions</i>		

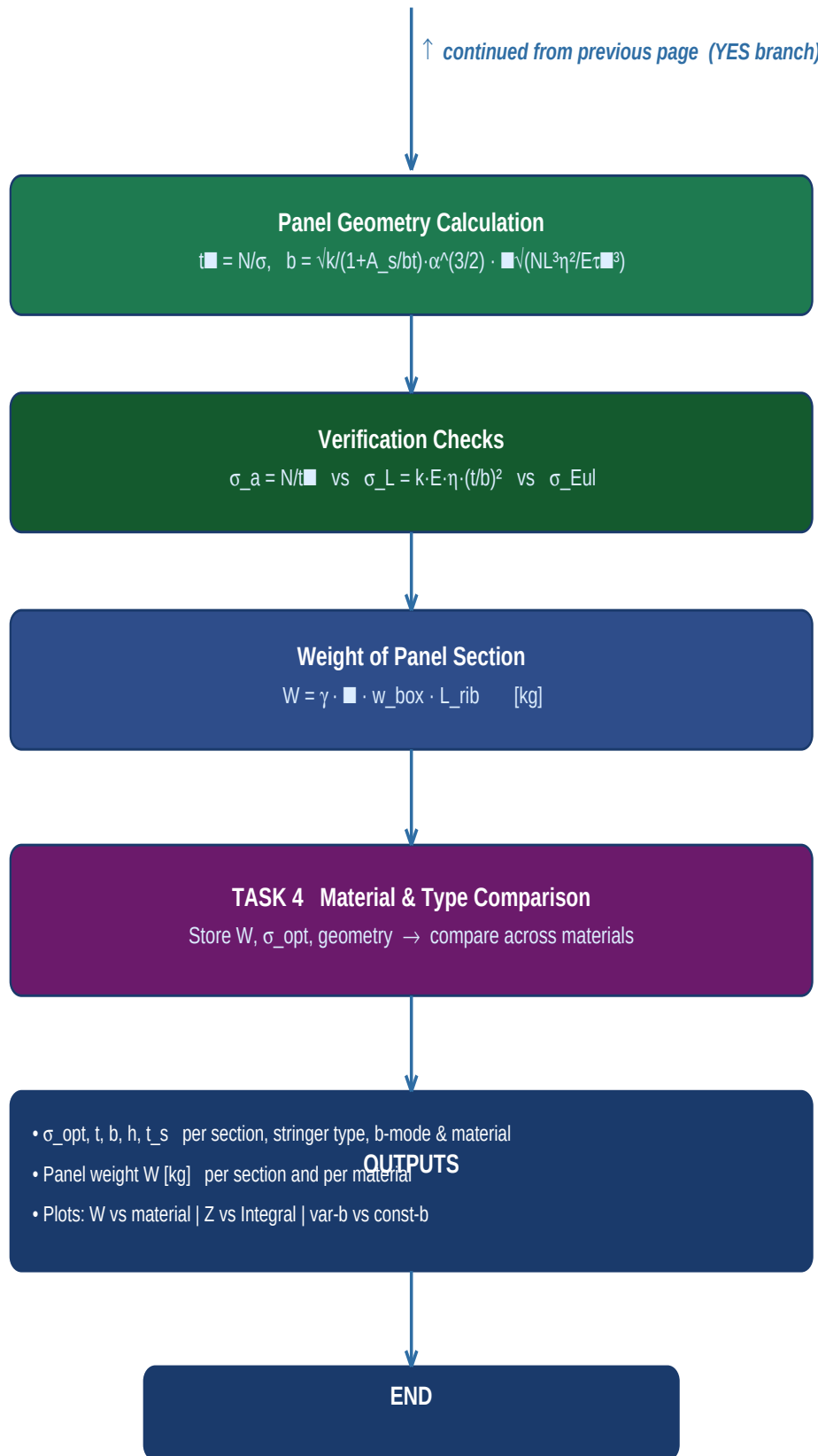
■ Workload Balance Summary

Person	Role	Tasks	Est. % effort
Person 1	Preliminaries & Geometry	1-I, 1-II, 1-III + shared input file	~20 %
Person 2	Variable-b / Z-stringer	2-I (a,b,c) + Z result table	~20 %
Person 3	Variable-b / Integral stringer	2-II (a,b,c) + weight comparison	~20 %
Person 4	Constant-b (manuf. constraints)	3-I to 3-IV	~20 %
Person 5	Material sensitivity + integration	4-I, 4-II + final report assembly	~20 %

MATLAB IMPLEMENTATION FLOWCHART — PROJECT II

A330-200 | Al 7075-T6 bare + alternative materials | Z-stringer & Integral stringer





Each function below maps directly to a step in the flowchart. Column **Function name** shows the suggested name; **Input parameters** lists what to pass in; **Output / Purpose** describes what it returns or does.

Functions

Function name	Input parameters	Output / Purpose
ramberg_osgood	sigma, mat_struct	Et, Es, eta (=Es/E), tau (=Et/E) — Ramberg-Osgood plastic moduli
find_alpha_Z	As_bt, ts_t, d_h	alpha_opt — structural efficiency for Z-stringer (chart interpolation)
find_alpha_INT	As_bt, ts_t	alpha_opt — structural efficiency for integral stringer
find_alpha_b_Z	As_bt, ts_t, b_fixed, N, E, tau_bar	alpha_b — efficiency for Z-stringer at fixed stringer pitch b
find_alpha_b_INT	As_bt, ts_t, b_fixed, N, E, tau_bar	alpha_b — efficiency for integral stringer at fixed b
plastic_cycle	alpha, N, E, L_prime, mat	[sigma_opt, tau_bar] — converged optimal stress with plasticity
panel_geometry	sigma_opt, N, k, alpha, As_bt, ts_t, eta, tau_bar, L	[b, t, ts, h, z_CG, b_W] — full panel dimensions
panel_weight	gamma, t_bar, w_box, L_rib	W [kg] — weight of one panel section

Loop variables

Loop variable	Values	Scope
mat	{Al7075-T6, Al2024-T3, Al2090-T83, Ti-6Al-4V}	Outer loop — Task 4 material comparison
sec	1 (root), 2 (kink), 3 (tip)	Section loop — repeated for each wing-box cross-section
str_type	{"Z", "INT"}	Stringer type — Z-shaped or integral
b_mode	{"var", "const"}	b variable (Tasks 2) or b constant (Task 3)

★ Shared tasks require coordination between persons and should be reviewed jointly before final submission. All MATLAB scripts should be version-controlled and documented with inline comments referencing equation numbers from the project slides.